

A Modular Python-Based Framework for Stochastic Techno-Economic Assessment of Thermal Power Plant Performance

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Abstract—This paper presents a Python-based simulation framework for assessing the long-term performance and economic viability of thermal power plants under uncertainty. The framework combines Monte Carlo reliability modeling, simplified thermodynamic simulation, and financial analysis in a modular, extensible architecture. Component failures and repairs are modeled using exponential distributions to generate time-varying availability profiles. These drive an hourly time-step simulation of plant operations over the full project horizon. Output metrics including energy production, fuel use, downtime, and efficiency are used to calculate financial indicators such as annual cash flow, Net Present Value (NPV), and Levelized Cost of Energy (LCOE). The framework supports advanced analytics, including real option valuation and Bayesian updating of failure rates, to enhance investment decision-making. An immutable configuration system ensures full reproducibility, and results are visualized through an interactive dashboard that summarizes key performance indicators and component behavior. Designed for clarity, flexibility, and scalability, this tool enables power sector stakeholders to evaluate project risks, explore reliability impacts, and support strategic planning using simulation-driven insights. The project is available on : [GitHub](#)

Index Terms—Monte-Carlo Simulation, Reliability modeling, Stochastic simulation, Net Present Value (NPV), Time-to-failure modeling, Exponential distribution, Financial viability assessment, Levelized Cost of Energy (LCOE), Bayesian statistics, Energy systems modeling, Decision support systems, Thermal power plants

I. INTRODUCTION

The global energy landscape is undergoing a profound transformation, driven by the dual imperatives of decarbonization and energy security. As power systems transition from centralized, baseload-dominated architectures to decentralized, variable renewable energy (VRE) systems, the operational and financial paradigms for thermal power plants are shifting fundamentally. No longer acting solely as steady-state energy providers, thermal assets are increasingly required to provide flexibility, grid inertia, and peak-shaving services [1]. In this volatile environment, traditional deterministic methods for techno-economic analysis such as static Net Present Value (NPV) calculations are proving insufficient for capturing the

complex interplay between component reliability, market uncertainty, and strategic management flexibility [2].

The planning and operation of large-scale power generation assets are inherently fraught with uncertainty. Factors ranging from component degradation and fuel price volatility to fluctuating regulatory frameworks can significantly impact a project's lifecycle viability. Decision-makers require sophisticated tools that can model not just the thermodynamic processes but also the stochastic economic elements that define a plant's success or failure. While modern management accounting tools like Activity-Based Costing (ABC) and the Balanced Scorecard (BSC) have been proposed to enhance strategic monitoring in power plants [3], [4], there remains a gap in integrating these managerial frameworks with high-fidelity, probabilistic operational simulations.

Existing literature highlights the economic risks associated with large power projects. [5] demonstrated that cost overruns in thermal power projects frequently exceed 40% of the initial budget, necessitating the use of hybrid predictive-probabilistic models to better quantify risk. Similarly, [6] argued for the adoption of stochastic optimization and risk-informed capital budgeting, specifically using metrics like Conditional Value at Risk (CVaR) to manage asset portfolios in nuclear environments. These studies underscore a critical industry need: a shift from static, "average-case" planning to dynamic, risk-aware simulation.

Furthermore, the rise of Integrated Energy Systems (IES) necessitates models that can optimize across multiple energy vectors (heating, cooling, electricity). [7] and [8] have shown that intelligent optimization and system integration can significantly reduce generation costs and improve exergy efficiency. However, evaluating the economic feasibility of such complex systems requires a simulation capability that can account for the specific reliability characteristics of interconnected components, as failure in one subsystem often cascades through the financial model [9].

This paper presents a comprehensive, modular, and extensible simulation framework designed to address these challenges. Developed in Python, the framework integrates System

modeling, stochastic reliability engineering using exponential failure distributions, and detailed economic calculations into a unified Monte Carlo simulation engine. Unlike static models, this system generates thousands of unique operational lifecycles, providing a probabilistic outlook on Key Performance Indicators (KPIs) such as NPV and Levelized Cost of Energy (LCOE). Additionally, the framework incorporates advanced analytical modules, including Real Options Analysis (ROA) to value managerial flexibility and Bayesian inference for updating failure rate estimates, thereby offering a holistic tool for strategic planning and risk assessment in the modern power sector.

II. METHODOLOGY AND SYSTEM ARCHITECTURE

The proposed simulation framework is designed with a modular, extensible, and reproducible software architecture, enabling long-horizon stochastic techno-economic analysis of thermal power plants. The system integrates three primary layers: configuration, simulation, and analytics. These are then linked through a linear and transparent data flow. This modular separation ensures that thermodynamic modeling, reliability simulation, and economic evaluation remain independent yet interoperable components, improving maintainability and facilitating extensibility for future studies [10].

Figure 1 illustrates the high-level system architecture.

A. Overview of Modular Architecture

The framework follows a structured pipeline consisting of four sequential stages:

- 1) **Configuration and Initialization** – Simulation parameters, economic assumptions, component reliability characteristics, and thermodynamic settings are loaded into an immutable configuration object.
- 2) **Stochastic Reliability Modeling** – Monte Carlo-generated availability profiles are computed for each component using exponential time-to-failure and time-to-repair distributions.
- 3) **Time-Step Process Simulation** – For each reliability scenario, a deterministic physical simulation is performed in hourly increments over the plant lifetime.
- 4) **Economic and Advanced Analytics** – Operational results are translated into financial metrics such as annual cash flows, NPV, and LCOE, followed by optional real options valuation and Bayesian failure-rate updating.

This architecture ensures that changes in one subsystem (e.g., reliability assumptions) do not affect others (e.g., economic calculations), thereby adhering to software-engineering principles of separation of concerns and modularity. A similar modular philosophy has been successfully adopted in prior techno-economic power system studies [1], [7], [8].

B. Centralized Configuration System

At the core of the framework lies a centralized configuration mechanism implemented using Python's `dataclasses` with the `frozen=True` option, creating an immutable `SimulationConfig` object. This design guarantees:

- **Reproducibility:** Each simulation run is governed by a single, version-controlled configuration artifact.
- **Traceability:** Economic assumptions, reliability parameters (MTBF/MTTR), simulation horizon, fuel characteristics, and plant specifications are stored together.
- **Safety:** No parameter can be altered during execution, eliminating accidental state mutations.

All downstream modules receive the configuration object as a read-only dependency, ensuring deterministic behavior across repeated runs. Such reproducible parameterization is consistent with best practices in techno-economic modeling and risk analysis [2], [6].

C. Simulation Core Engine

The simulation core, implemented in `src/simulation_core.py`, orchestrates the entire workflow through the primary function `run_simulation()`. The engine performs three major operations:

1) *Plant Component Modeling Assumptions:* To maintain computational tractability and modular extensibility during early experimentation, all plant components are modeled using simplified first-order transfer functions. Each subsystem: boiler, turbine, condenser, pump, heat exchanger is treated as a first-order lag system with fixed response parameters. This assumption balances physical realism with computational performance, making the framework suitable for large-scale Monte Carlo simulations across extended time horizons. Formally, the output of each component is governed by:

$$\tau \frac{dy(t)}{dt} + y(t) = K \cdot u(t), \quad (1)$$

where:

- $y(t)$ is the output state (e.g., temperature, pressure, flow rate),
- $u(t)$ is the input stimulus or control signal,
- K is the component gain (steady-state response),
- τ is the time constant representing component inertia.

This abstraction reduces simulation complexity while preserving essential dynamics. It also enables modular tuning of each component via the configuration file. As the framework matures, higher-order models or physics-informed surrogate models can be substituted without architectural changes.

2) *Monte Carlo Reliability Simulation:* Component reliability is modeled using exponential probability distributions for both Time-to-Failure (TTF) and Time-to-Repair (TTR). For each component i , random failure and repair durations are generated using:

$$\text{TTF}_i = -\text{MTBF}_i \cdot \ln(U_1), \quad \text{TTR}_i = -\text{MTTR}_i \cdot \ln(U_2),$$

where U_1 and U_2 are uniformly distributed random variables in $[0, 1]$. This approach aligns with standard stochastic reliability literature and mirrors models used in risk-based predictive-probabilistic analyses of power plant systems [5], [9].

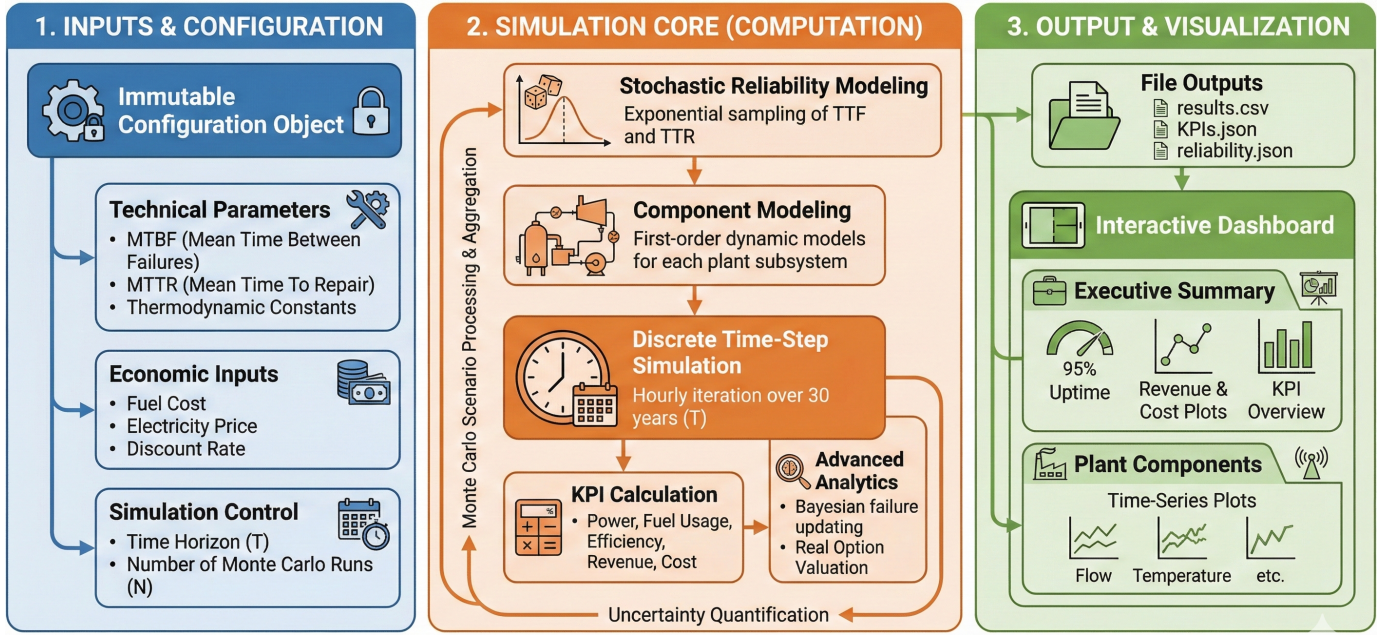


Fig. 1. Simulation Architecture

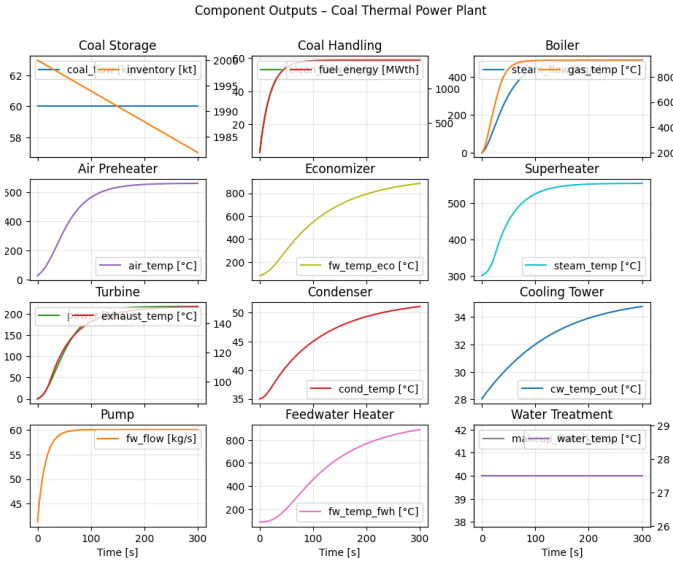


Fig. 2. Plant Components Modeled as a First order System with Lag as necessary

A binary availability vector $A_i(t)$ is constructed for each hour of the simulation horizon, marking the operational state of each component. Running multiple Monte Carlo iterations yields diverse reliability scenarios capturing variability in plant behavior.

3) *Time-Step Thermodynamic Simulation*: For each availability scenario, the engine runs a deterministic time-step simulation via the `simulate_process()` function. At each

hourly interval:

- 1) Component availability is checked via the reliability profile.
- 2) Plant thermodynamic outputs (power generation, heat rate, efficiency) are computed.
- 3) Fuel consumption, energy losses, and downtime hours are recorded.

This hybrid architecture's stochastic availability combined with deterministic physics allows the framework to produce realistic long-horizon performance trajectories. Similar hybrid modeling approaches have been applied in integrated system optimization and flexibility retrofit studies [1], [8].

4) *KPI and Economic Computation*: Raw time-series data are passed to economic modules (`economic_models.py`) to compute:

- Annual and cumulative revenues
- Fuel, maintenance, operational, and capital expenditures
- Net Present Value (NPV)
- Levelized Cost of Energy (LCOE)
- Profitability and payback period

The economic computation follows discounted cash flow (DCF) principles widely used in economic feasibility studies of thermal power plants [2]. Additional advanced analytics include:

- **Real Option Valuation** – An adapted Black–Scholes model for investment deferral.
- **Bayesian Failure-Rate Updating** – Incorporating new reliability observations into posterior estimates, consistent with modern reliability engineering practices.

D. Data Storage, Versioning, and Visualization

All results are stored in a structured directory under `sim_out/`, including:

- Time-series operational data (`results.csv`)
- Economic KPI summaries (`KPIs.json`)
- Component reliability metrics (`reliability.json`)

An interactive dashboard built with Plotly Dash provides visual analysis of:

- 1) Component-level reliability behavior
- 2) Executive economic summaries
- 3) Energy and fuel trends across Monte Carlo scenarios

This visualization layer functions as a decision-support interface, enabling intuitive navigation of high-dimensional simulation outputs similar to the decision-support systems discussed in strategic management and BSC-driven performance frameworks [3], [4].

III. MATHEMATICAL MODELING

A. Stochastic Reliability Models

The reliability behaviour of each plant component is modeled stochastically using an Exponential Distribution, a standard formulation in reliability engineering due to its memoryless property. This assumption implies that the probability of failure is independent of a component's age, making it suitable for high-volume operational equipment in thermal power environments [5], [9].

The simulation implements inverse transform sampling to generate failure and repair durations. For each component, the Time to Failure (T_{fail}) and Time to Repair (T_{repair}) are computed as:

$$T_{\text{fail}} = -MTBF \cdot \ln(U_1) \quad (2)$$

$$T_{\text{repair}} = -MTTR \cdot \ln(U_2) \quad (3)$$

where:

- $MTBF$ is the Mean Time Between Failures.
- $MTTR$ is the Mean Time To Repair.
- $U_1, U_2 \sim \mathcal{U}(0, 1)$ are independent uniformly distributed random variables.

Repeated sampling across multiple Monte Carlo iterations yields a set of binary availability profiles for each component, which drive the system-level time-step simulation. This hybrid reliability-physics structure mirrors probabilistic risk-modelling techniques applied in recent thermal and nuclear project assessments [5], [6].

B. Economic Formulations

Operational time-series outputs are translated into economic metrics to quantify annual profitability and long-term project viability. The annual profit π_{annual} is computed as:

$$\pi_{\text{annual}} = R_{\text{energy}} - (C_{\text{fuel}} + C_{\text{fixed}} + C_{\text{variable}}) \quad (4)$$

Revenue and fuel cost components are given by:

$$\begin{aligned} R_{\text{energy}} &= E_{\text{out}} \times P_{\text{elec}}, \\ C_{\text{fuel}} &= M_{\text{fuel}} \times P_{\text{fuel}}, \end{aligned} \quad (5)$$

where:

- E_{out} is the annual energy output (MWh),
- P_{elec} is the electricity price (\$/MWh),
- M_{fuel} is the annual fuel consumption (tons),
- P_{fuel} is the price per ton of fuel.

These economic calculations align with standard discounted cash-flow-based viability assessments performed in earlier thermal power plant case studies [2].

C. Advanced Analytics and KPIs

Long-term financial feasibility is assessed using Net Present Value (NPV), which discounts future cash flows to reflect risk and the time value of money:

$$NPV = \sum_{t=0}^N \frac{CF_t}{(1+r)^t} - I_0, \quad (6)$$

where:

- CF_t is the net cash flow in year t ,
- r is the discount rate,
- N is the project lifetime (years),
- I_0 is the initial capital investment.

NPV provides a single, risk-adjusted indicator summarizing profitability and is widely used in economic evaluations of power generation projects [2], [6].

Beyond DCF metrics, the framework incorporates:

- **Real Option Analysis (ROA)** using an adapted Black-Scholes formulation to quantify the financial value of deferring investment under uncertainty, reflecting modern risk-informed project evaluation techniques [6].
- **Bayesian Updating of Failure Rates**, where the posterior estimate of the failure rate λ is recalculated as new operational or simulated reliability observations become available. This allows the framework to represent learning effects and evolving asset conditions [10].

These advanced analytics extend the framework's capability beyond static evaluations, enabling more realistic modeling of investment flexibility and dynamic reliability behaviour.

IV. RESULTS AND VISUALIZATION

The simulation framework produces a comprehensive set of operational, reliability, and economic outputs that collectively describe the performance of the power plant across all Monte Carlo scenarios. These outputs are systematically stored, structured, and visualized through an interactive dashboard designed to support analytical decision-making. This section presents the organization of the data generated by the framework as well as the visualization tools used to interpret long-horizon simulation results.

A. Data Output Structure

Each simulation run generates a dedicated results directory within the `sim_out/` folder, ensuring clear separation and reproducibility across experiments. The output consists of three primary data categories:

- 1) **Operational Time-Series Data** Stored in `results.csv`, this dataset captures hourly performance metrics across the entire simulation horizon. Key variables include:

- Electrical power output,
- Component availability indicators,
- Fuel consumption rates,
- Heat rate and efficiency trends,
- Downtime events and durations.

These time-series allow detailed analysis of operational reliability and performance variability across Monte Carlo scenarios.

- 2) **Reliability Metrics** The file `reliability.json` summarizes component-level reliability statistics such as:

- Mean availability,
- Failure counts,
- Average downtime per failure,
- Total outage duration across the lifetime.

These metrics quantify the stochastic behavior of each subsystem, revealing critical bottlenecks and opportunities for maintenance optimization.

- 3) **Economic and Financial Indicators** Stored in `KPIs.json`, these outputs consolidate year-wise and cumulative economic performance, including:

- Annual revenue,
- Fuel and operating costs,
- Discounted cash flows,
- Net Present Value (NPV),
- Levelized Cost of Energy (LCOE),
- Profitability metrics such as payback period.

This structured output format allows seamless integration with sensitivity studies, scenario comparison, or policy evaluation analyses.

Additionally, metadata files store versioning information, configuration parameters, timestamps, and random seeds, enabling the framework to reproduce any simulation run exactly. This careful design supports best practices in computational experiment reproducibility and model transparency.

B. Interactive Dashboard

To support high-level interpretation of the simulation outputs, the framework provides an interactive dashboard built using Plotly Dash. The dashboard serves as the analytical front-end of the system, enabling users to explore operational behavior, reliability characteristics, and economic performance across the entire simulation horizon. It offers intuitive visual summaries as well as detailed subsystem-level plots, making the results accessible to engineers, analysts, and decision-makers.

The dashboard is structured into two main tabs: the **Executive Summary** and the **Plant Components** view.

1) **Executive Summary View:** The Executive Summary tab consolidates the key techno-economic indicators of a selected simulation run. At the top of the view, metadata such as the run folder, timestamps, and total runtime are displayed. The main content consists of a grid of key performance indicators (KPIs), including:

- Total energy generated (MWh),
- Capacity factor (%),
- Cost per MWh and total revenue,
- O&M cost, net profit, and profit margin,
- Uptime percentage and downtime hours,
- MTBF and aggregated failure statistics,
- Net Present Value (NPV) and investment metrics,
- Profit VaR and Profit CVaR (financial risk measures),
- Weibull distribution parameters for reliability analysis,
- Bayesian-updated MTBF.

These indicators provide an immediate assessment of the financial viability, operational stability, and risk exposure of the plant over the simulated lifetime.

Below the KPIs, the dashboard displays several high-level plots for financial and operational interpretation. Placeholders for these figures are included below:

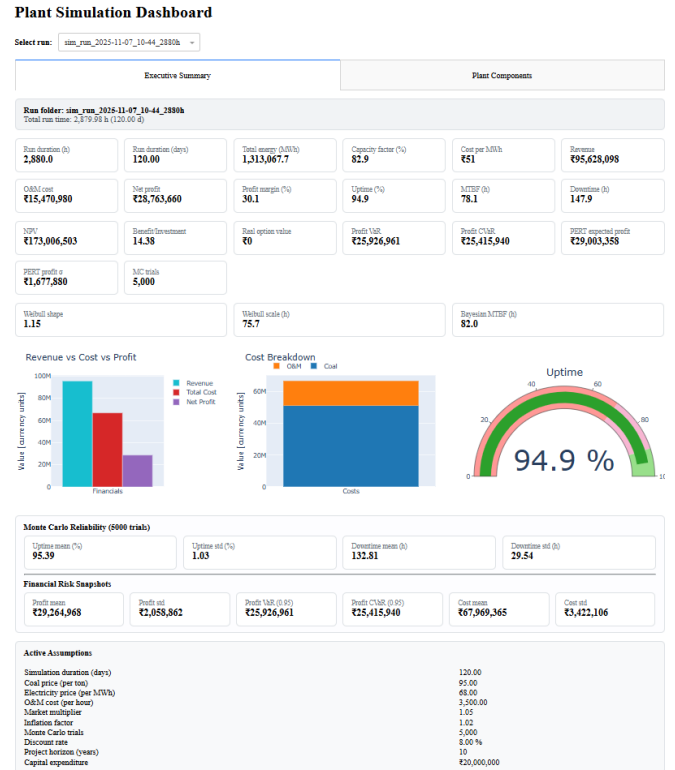


Fig. 3. Executive Summary view displaying KPIs, revenue vs. cost breakdown, uptime gauge, and reliability statistics.

A dedicated section titled *Active Assumptions* lists all economic and simulation parameters used for the run, ensuring

complete transparency and reproducibility of results.

2) *Plant Components View*: The Plant Components tab provides detailed time-series visualizations of subsystem behavior over the full simulation horizon. This view displays hourly trends for major thermodynamic and process variables, including:

- Water, clean, and steam flow rates,
- Fuel energy input,
- Boiler gas and steam temperatures,
- Air preheater and exhaust temperatures,
- Condenser and cooling water temperatures,
- Feedwater flow and heater outlet temperatures,
- Reliability trend signals.

These plots help identify operational disturbances, equipment stress patterns, and failure-induced performance variations. The consistent scaling and color-coded legends support direct visual comparison across components.

A placeholder for the component-level visualization figure is provided below:

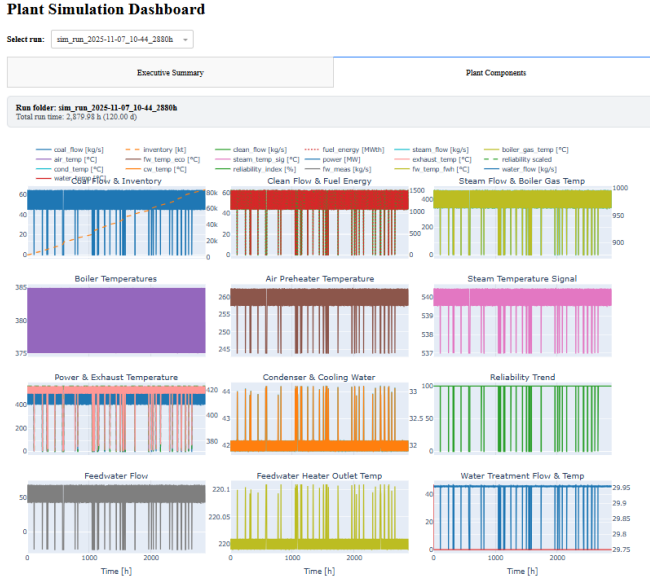


Fig. 4. Plant Components view showing time-series behavior of key thermal and reliability variables.

Interactive controls such as scenario selectors, drop-down menus, and hover-to-inspect tooltips allow the user to examine specific Monte Carlo scenarios, zoom into periods of interest, and study subsystem-level fluctuations in detail.

Overall, the interactive dashboard transforms high-dimensional simulation outputs into an intuitive analytical environment. By integrating real-time visualization, structured KPIs, and detailed component trends, it enables comprehensive exploration of the reliability–economics interactions driving long-term plant performance.

V. CONCLUSION AND FUTURE WORK

This paper introduced a modular Python-based simulation framework for stochastic techno-economic assessment of power plant performance. By integrating Monte Carlo reliability modeling, first-order thermodynamic simulation, and economic analysis, the system captures the coupled dynamics of component behavior, operational uncertainty, and financial viability. The use of an immutable configuration system ensures reproducibility, while an interactive dashboard enhances result interpretation and decision support.

Future development will focus on incorporating more advanced reliability models (e.g., Weibull distributions), expanding thermodynamic detail, and modeling fuel price uncertainty. Additional directions include optimization features (e.g., retrofit planning) and broader application to integrated or renewable–thermal hybrid systems.

VI. LIMITATIONS

The current framework simplifies component behavior using first-order models and assumes exponentially distributed failure and repair times. External influences such as electricity market dynamics, regulatory constraints, and emissions impacts are not yet modeled. Additionally, validation against real-world operational data is pending, and the current analysis focuses on a single plant configuration without scenario comparison or sensitivity analysis.

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